

Day 12: Conditional Probability and Bayes' Theorem

Learning How Probabilities Change with Information

100 Days of Data Science

Why Conditional Probability is Important

In real life, probabilities change when we gain new information.

Machine Learning models constantly update predictions based on new data.

Conditional probability helps answer:

- What is the chance of an event happening given something already happened?

What is Conditional Probability?

Conditional probability is the probability of event A happening given that event B has occurred.

It is written as:

$$P(A|B)$$

Which means: probability of A given B.

Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Simple Example

Suppose:

- 60% students passed Math
- 30% passed both Math and Science

Probability of passing Science given student passed Math:

$$P(\textit{Science}|\textit{Math}) = \frac{0.30}{0.60} = 0.5$$

Real-Life Example: Rain and Traffic

If it rains, chances of traffic jam increase.

So:

- $P(\text{Traffic})$ without rain is low
- $P(\text{Traffic} \mid \text{Rain})$ is high

This is conditional probability in daily life.

Why ML Uses Conditional Probability

Machine Learning constantly asks:

- What is probability of class given features?

Example:

$$P(\textit{Spam}|\textit{Email words})$$

This means: Probability email is spam given the words inside it.

Introduction to Bayes' Theorem

Bayes' Theorem helps reverse conditional probabilities.

It answers:

- What is probability of cause given the effect?

Bayes Formula

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

- $P(A)$ = prior probability
- $P(B|A)$ = likelihood
- $P(B)$ = evidence

Medical Diagnosis Example

Suppose:

- 1% people have a disease
- Test detects disease 99% correctly

If test is positive, Bayes helps calculate actual chance of having disease.

Without Bayes, doctors may overestimate risk.

Bayes in Machine Learning (Naive Bayes)

Naive Bayes classifier works fully on Bayes Theorem.

It calculates:

$$P(Class|Features)$$

Used in:

- Spam detection
- Sentiment analysis
- Document classification

Real Example: Spam Filter

Model checks:

- Probability email is spam given words like "free", "offer", "win"

Then predicts class with highest probability.

Why Bayes is Powerful

- Works with small data
- Fast and efficient
- Handles uncertainty well

Common Mistakes

- Ignoring prior probabilities
- Confusing $P(A \rightarrow B)$ with $P(B \rightarrow A)$
- Assuming probabilities mean certainty

Final Takeaway

Conditional probability updates beliefs using new information.

Bayes' Theorem is the backbone of many AI systems.

Understanding this gives you deep insight into how Machine Learning thinks.